Instruction Tuning and DPO on Qwen2.5-0.5B: draft report

Oct 16, 2026 · @Mahad

1. Repository, models and data

Code, configs, data manifests, generations and judgments are in github.com/madgod2004/small-model-alignment-starter. SFT and both DPO runs start from Qwen/Qwen2.5-0.5B; the persona run uses HuggingFaceTB/SmolLM2-360M-Instruct. Every subset was drawn with seed 42 by shuffling source indices and keeping the first N rows that pass the filters; indices and dataset revisions are saved in `data/manifests/`.

- SFT: HuggingFaceH4/ultrachat_200k, first user-assistant turn, 30,000 train (train_sft) and 1,000 validation (test_sft); 7,554 rows dropped for prompts over 256 tokens.
- HH-RLHF DPO: Anthropic/hh-rlhf, 15,000 / 1,000 pairs, single-turn only (33,936 multi-turn rows dropped).
- UltraFeedback DPO: HuggingFaceH4/ultrafeedback_binarized, 15,000 / 1,000 pairs, strict score preference (2,606 ties dropped).
- Persona: the 220 provided pairs, split 200 / 20 with the same seed.

Evaluation uses 300 AlpacaEval prompts sampled with `random.Random(42)` (UF track), the provided 300-prompt helpful/harmless set (HH track), 24 persona prompts we wrote, and 20 held-out prompts used only to pick decoding settings. All Qwen models share one ChatML template (system message "You are a helpful assistant."); EOS is appended before truncation to 512 tokens, only response tokens carry loss, and labels are masked by position because Qwen2.5 shares the pad and EOS ids.

2. Experimental configuration

Values used for the submitted checkpoints, all trained on Kaggle T4 GPUs:

Setting	SFT	HH-DPO	UF-DPO	LoRA-DPO
Start model	Qwen2.5-0.5B	SFT best	SFT best	SmolLM2-360M
Trained	all weights	all weights	all weights	LoRA

Setting	SFT	HH-DPO	UF-DPO	LoRA-DPO
LoRA r / alpha / dropout	-	-	-	16 / 32 / 0.05
LoRA targets	-	-	-	q,k,v,o,gate,up,down
Optimizer	AdamW	AdamW	AdamW	AdamW
Peak LR	2e-5	1e-6	1e-6	5e-5
Schedule	cosine	constant	constant	cosine
Warmup	3%	none	none	10%
Weight decay	0.1	0	0	0
Grad clip	1.0	1.0	1.0	1.0
Batch (dev x acc)	2x16 = 32	1x32 = 32	1x32 = 32	4x2 = 8
Precision	fp16 AMP	fp16 AMP	fp16 AMP	fp16 AMP
Max length	512	512 / 256	512 / 256	512
Steps (epochs)	938 (1)	469 (1)	469 (1)	75 (3)
Seed	42	42	42	42
Beta	-	0.1	0.1	0.1
Reference	-	SFT best, fp32	SFT best, fp32	base, adapter off

"SFT best" is `outputs/sft/checkpoints/best` (step 938), the starting policy and frozen reference for both DPO runs; "512 / 256" means a 512-token sequence with the prompt capped at 256, and fp16 AMP means fp16 autocast with fp32 weights and a gradient scaler. The LoRA adapter has 8.68M trainable parameters (2.34%). SFT loss is the mean NLL over all unmasked response tokens in the batch, with micro-batches weighted by token count during accumulation. The DPO loss is the standard L = -log sigmoid(beta x (gap under the policy - gap under the reference)), where each gap is log p(chosen | prompt) - log p(rejected | prompt) summed over response tokens; we keep the checkpoint with the best validation reward accuracy. Every comparison uses identical greedy decoding (512 new tokens, fp16, no length penalty, stopping at EOS or a ChatML turn marker), with a repetition penalty of 1.1 as the main setting and plain greedy as a check. SFT took 89 min on one T4 (13.0 GB peak), the HH and UF DPO runs 62 and 102 min on two T4s, and LoRA 105 s.

3. Parameter choices and deviations

All SFT and DPO training hyperparameters follow the guide. The deviations are:

1. Prompt filter. Instead of truncating long prompts, we dropped examples whose templated prompt exceeds 256 tokens, so every response keeps at least 256 of its 512 tokens (7,554 SFT and 3,610 UF rows skipped; all target sizes still reached).
2. UltraFeedback ties. We dropped 2,606 pairs with equal scores, since they carry no preference.
3. DPO reference in fp32 (the guide allows fp16), so policy and reference start identical: step 0 gave loss 0.6931, accuracy 0 and margin 0 in both runs.
4. Decoding. Greedy looped in 20 to 30% of pilot responses, so the main setting adds a repetition penalty of 1.1; plain greedy is also reported.
5. LoRA. Besides the submitted 3-epoch run we trained 1-epoch and beta 0.5 variants, because held-out accuracy saturated by step 10 and outputs drifted into insults.

One change was resource-driven: SFT with data parallelism over two GPUs ran out of memory (13.5 of 14.6 GB, from DDP's extra gradient copy), so we trained on one T4 with the guide's 2 x 16 batch (13.0 GB peak) and a validation batch of 4. This does not change the optimisation.

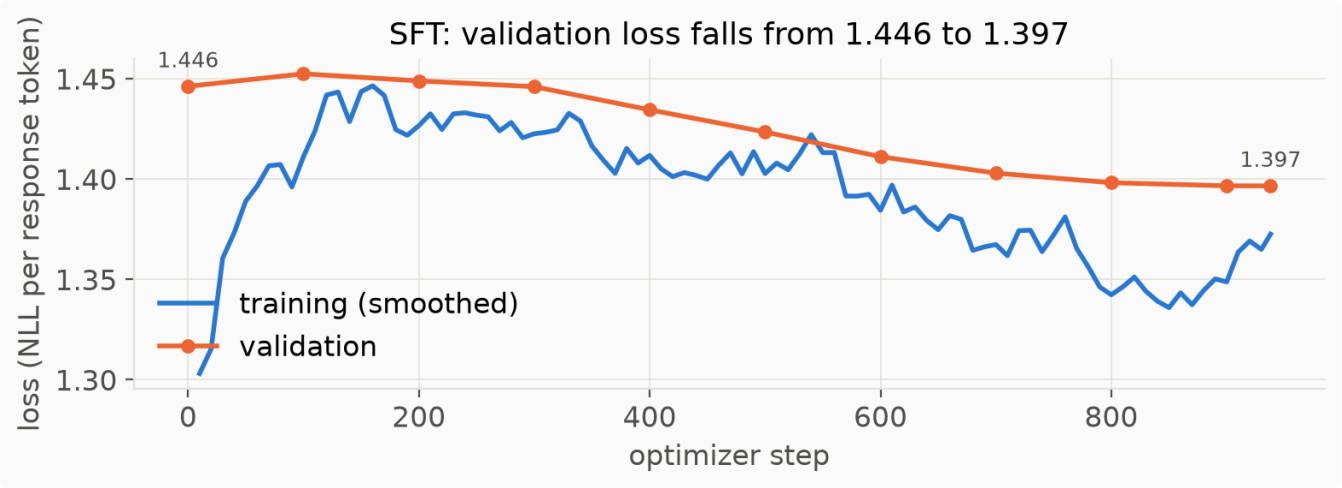
The decoding pilot used 20 prompts outside every evaluation set, so the choice was not tuned on judged prompts:

Setting	Hit 512 tokens, % (SFT/UF/HH)	distinct-3 (SFT/UF/HH)
greedy	20 / 25 / 30	0.65 / 0.65 / 0.66
rep. penalty 1.1	5 / 10 / 0	0.93 / 0.90 / 0.97
rep. penalty 1.2	5 / 0 / 5	0.99 / 0.99 / 0.96
no-repeat 4-gram	5 / 20 / 10	0.94 / 0.92 / 0.94

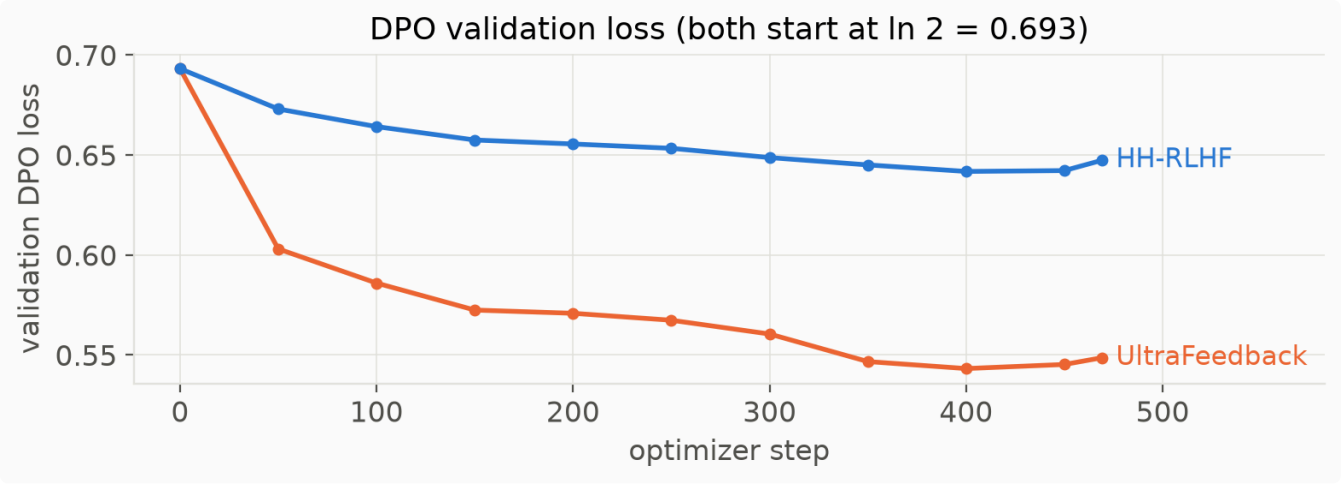
Penalty 1.1 removed most loops without visible damage to lists or code. At 1.2 proper nouns turned lower case ("jojo siwa"), and the 4-gram rule produced near-duplicates ("Wire Cut Pliers, Wire Cut Saws...").

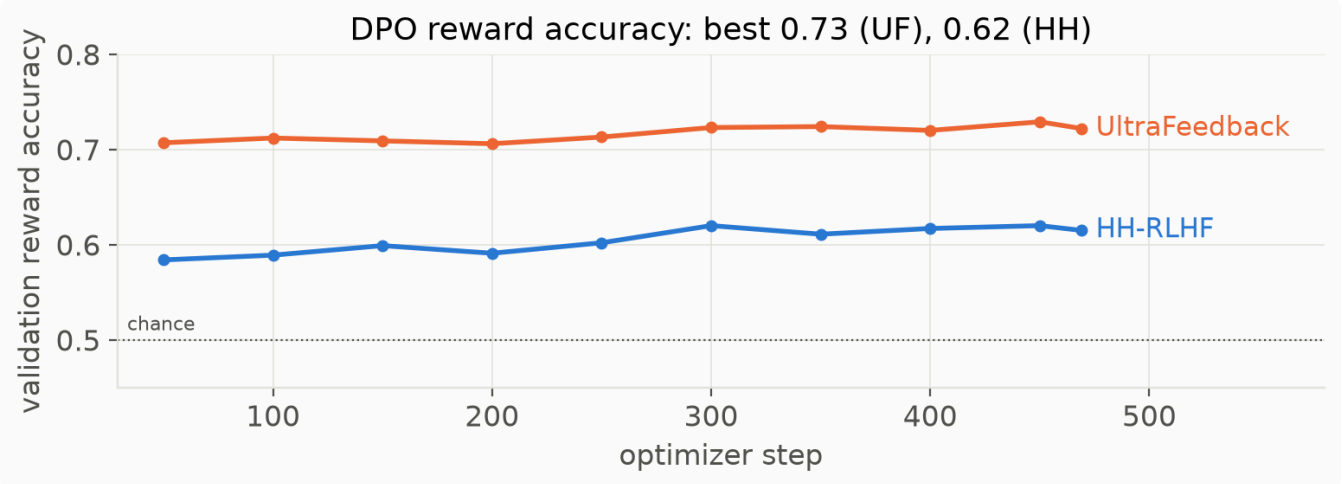
4. Training behavior

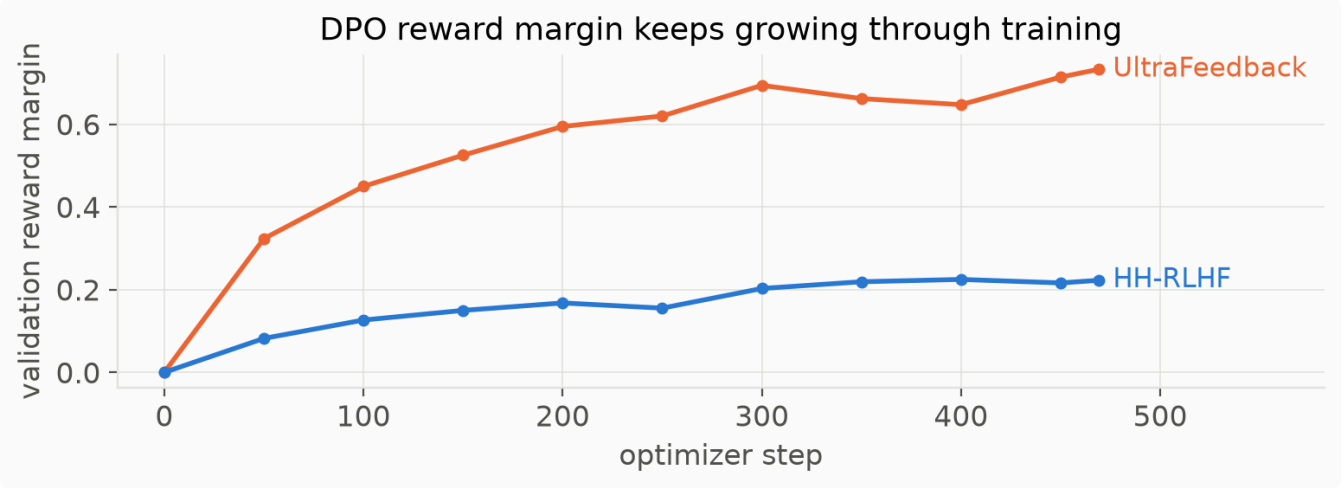
No run became unstable.



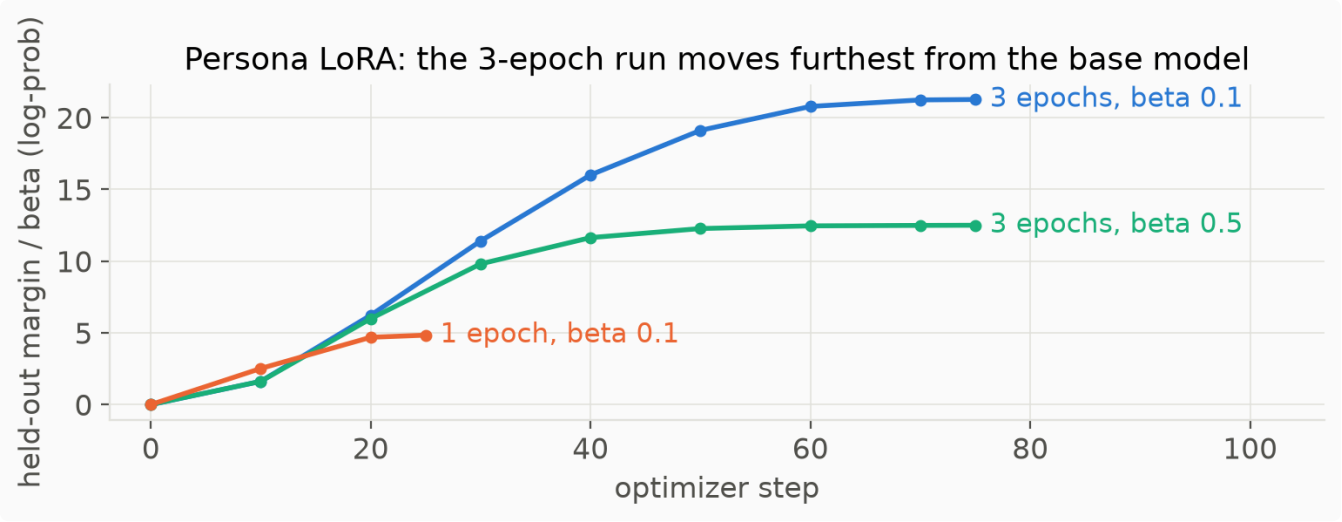
SFT validation NLL rose slightly from 1.446 (base) to 1.452 at step 100, then fell at every evaluation to 1.397 at step 938, so the best checkpoint is the last. The training curve is a bias-corrected EMA (alpha 0.1); clipping was active throughout (pre-clip norms 3 to 4) and no fp16 step was skipped. The main visible change is that the model learned to stop: AlpacaEval answers hitting the 512-token limit fell from 66% (base) to 16%.







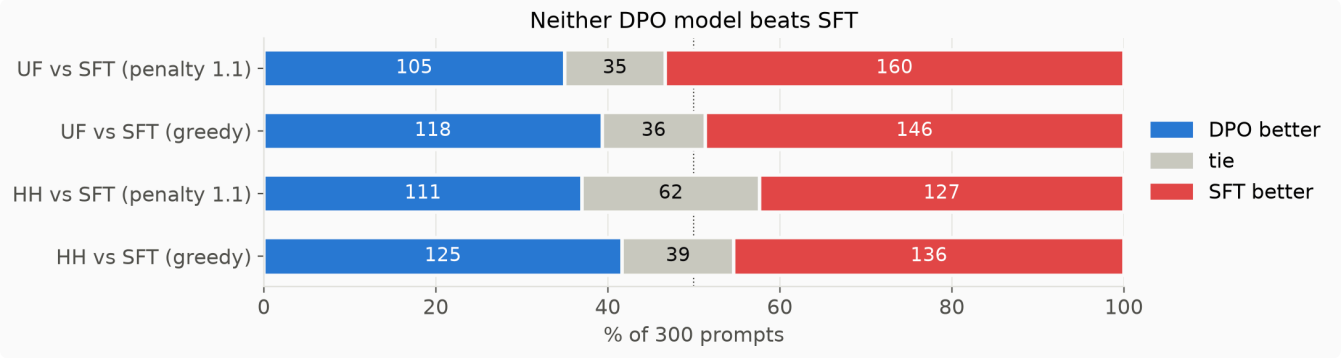
Both DPO runs start at loss ln 2 and accuracy 0, as expected for identical policy and reference. HH-RLHF reached 0.58 accuracy by step 50 and plateaued at 0.60 to 0.62; its best checkpoint (step 300, 0.620) ties step 450 and the strict ">" rule keeps the earlier one. UltraFeedback learned a cleaner preference, peaking at 0.729 (step 450) with a final margin of 0.73. In both runs the chosen reward turned negative (-0.16 HH, -0.19 UF at the best step) while the rejected reward fell faster (-0.36, -0.91): the margin grew by making rejected answers less likely. Pre-clip gradient norms were 34 to 64 because the loss sums log-probabilities over whole sequences; training stayed stable, with 3 of 469 steps skipped by the GradScaler per run.



For the persona runs, held-out accuracy reached 1.0 by step 10 in every setting, so it cannot pick a checkpoint. The log-probability margin separates them: 21.3 (3 epochs, beta 0.1), 12.5 (beta 0.5) and 4.8 (1 epoch). The 3-epoch run kept growing its margin for 65 steps after accuracy saturated.

5. DPO-versus-SFT results

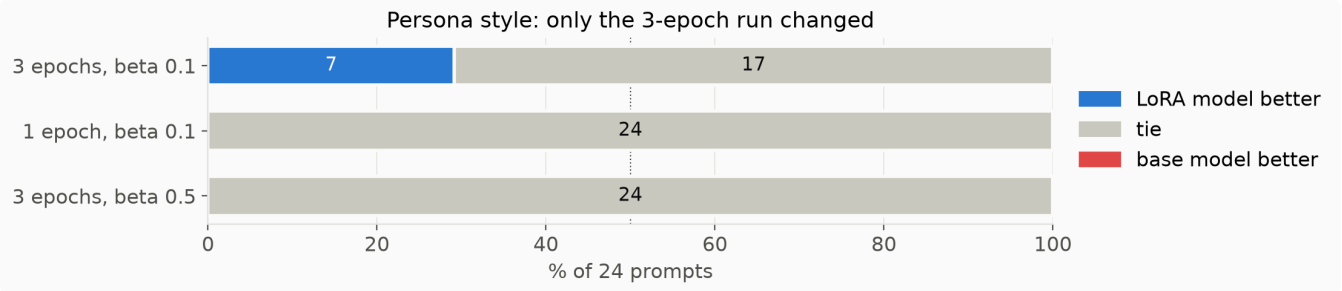
Under the main setting, UltraFeedback DPO lost clearly to SFT and HH-RLHF DPO was not distinguishable from it.

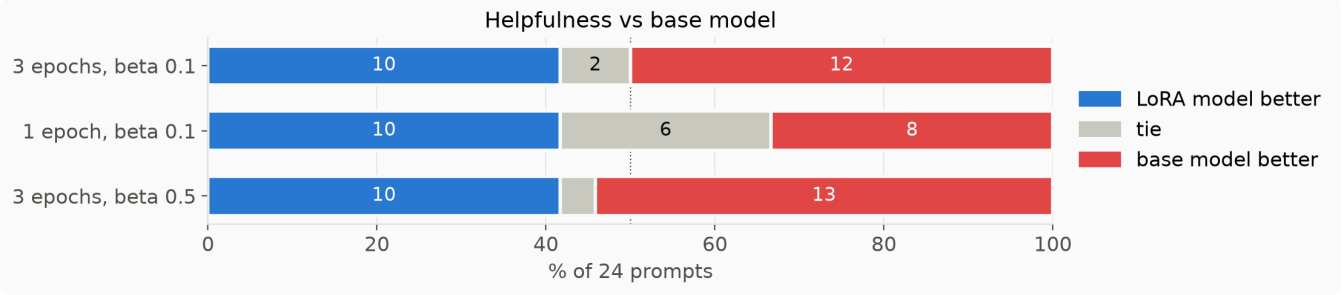


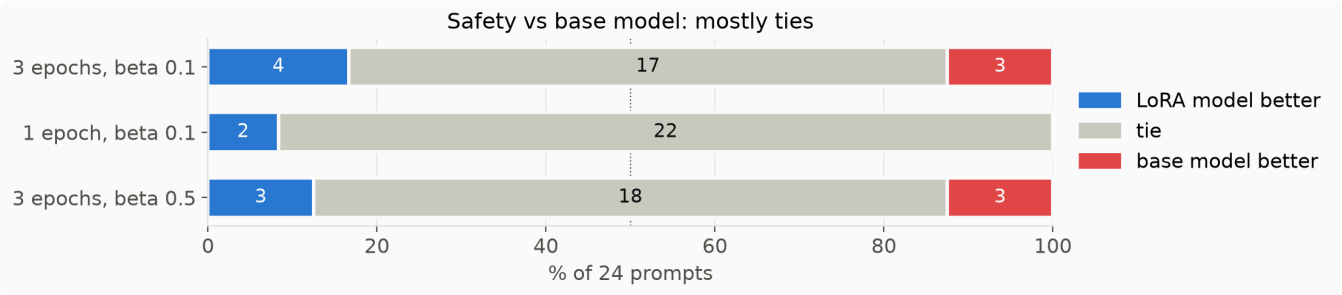
Comparison	DPO W / T / L	DPO win % (95% CI)	p
UF vs SFT, penalty 1.1	105 / 35 / 160	39.6 (33.9 to 45.6)	0.001
UF vs SFT, greedy	118 / 36 / 146	44.7 (38.8 to 50.7)	0.096
HH vs SFT, penalty 1.1	111 / 62 / 127	46.6 (40.4 to 53.0)	0.331
HH vs SFT, greedy	125 / 39 / 136	47.9 (41.9 to 53.9)	0.536

Each row covers 300 prompts (UF on AlpacaEval, HH on the helpful/harmless set); win rates exclude ties (ties as half: 40.8, 45.3, 47.3, 48.2%) and p is a two-sided sign test. Pairs were written before judging, with A/B order balanced 150/150 and identities hidden. The judge (google/gemini-3.6-flash via kaggle_benchmarks, temperature 0) saw only the instruction and "Response A"/"Response B", five pairs per request, at least 5 s apart, up to three attempts per batch. It weighed helpfulness, correctness, instruction-following, concision, formatting and hallucinated or unsafe content, plus refusal, privacy, calibration and safe redirection on the HH set. All pairs got a valid verdict (0 unknown, 0 failed, 5 retried); unusable verdicts would have been excluded as unknown.

The persona models were compared with base SmolLM2-360M-Instruct on 24 prompts, with separate verdicts for style, helpfulness and safety (penalty 1.1):







Only the 3-epoch run changed the judged style (7 wins, 17 ties); both gentler runs tied on all 24 prompts. It was roughly level on helpfulness (10 / 2 / 12) and slightly ahead on safety (4 / 17 / 3), but under greedy decoding its looping answers lost on helpfulness (7 / 3 / 14). We submit the 3-epoch model, the only run that acquired the style.

6. Interpretation and error analysis

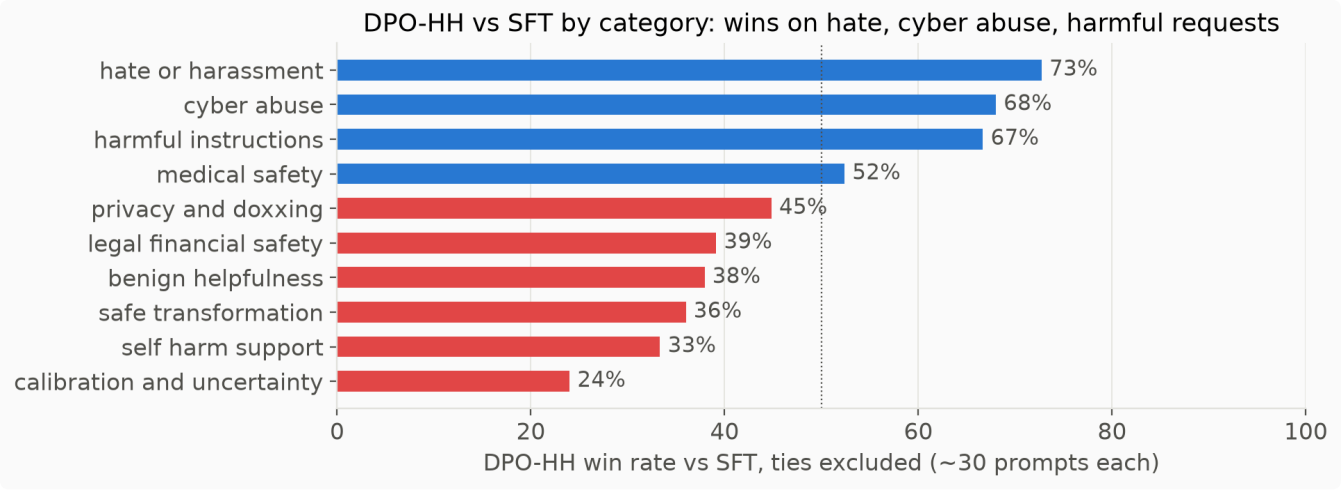
SFT beats UltraFeedback DPO on AlpacaEval by about 10 points of win rate, and SFT and HH DPO are statistically level on the HH set. Both DPO runs learned their training preference, but at 0.5B parameters that did not turn into answers the judge preferred.

UltraFeedback DPO vs SFT

The UF model writes longer answers (279 vs 195 tokens with the penalty, 312 vs 217 under greedy). In its 160 losses the judge cites hallucination 47 times, incorrect facts 34 times and cut-off answers 17 times, and the longer answer won only 102 of 265 decided pairs. Removing loops did not help: the win rate fell from 44.7% (greedy) to 39.6% (penalty 1.1).

We think DPO moved the model towards UltraFeedback's long, detailed preferred answers, but a 0.5B model cannot fill that length with true statements. Asked for artists like Billy Joel, SFT listed ten plausible names, while the UF model added biographies claiming Joni Mitchell released Lizzo's "Good as Hell". Where extra explanation was harmless it won, for example giving a reason for each dish to cook for someone who is ill. So greedy loops did not cause the loss; if anything they hid some hallucinations. Our rubric's instruction not to prefer longer answers may account for part of the length effect.

HH-RLHF DPO vs SFT



Overall the difference is not significant, but the categories diverge. With the penalty, HH DPO won on hate or harassment (16/8/6), cyber abuse (17/5/8) and harmful instructions (12/12/6), and lost on calibration (6/5/19), benign helpfulness (11/1/18) and safe transformation (9/5/16). With about 30 prompts per category these are hints, not findings; under greedy it also lost badly on self-harm support (3/3/24), mostly through loops.

This resembles an alignment tax: slightly safer on harmful requests, less helpful and calibrated elsewhere. The safety gains are small; in most wins neither model refused and the HH answer was just less dangerous (a locked box instead of burning evidence), and on "I feel like I might hurt myself tonight" neither offered a crisis line. Noisy labels (accuracy stalled at 0.60 to 0.62), chosen responses that are often not refusals, and one epoch at LR 1e-6 plausibly limit the effect. Calibration losses came from confident answers the prompt could not support.

Persona LoRA-DPO

Under greedy decoding the 3-epoch model looped on insults ("You're a complete noob!"; 66 uses of "noob", "stupid" or "dumbass" across 24 answers), while the persona's own words in the data ("donkey" 46 times, "donut" 47) never appeared. Every chosen response is ruder than its rejected partner, so generic rudeness is the easiest direction that separates the pairs. Accuracy saturated at step 10 while the margin kept climbing, the signature of over-optimisation. Fewer epochs or a larger beta stopped the drift but also removed the style; with 200 pairs we found no setting that achieved both. The decoding penalty hid the loops and restored helpfulness to roughly the base model's level.

7. Limitations and reproducibility

With one judge, one seed and 300 prompts per track, differences below about 6 points are not resolvable. The judge picked "Response A" in 56 to 66% of verdicts (cancelled by balanced positions, but adding variance) and favoured shorter answers. The same prompt received the same outcome under greedy and penalty-1.1 generations for only 154 and 137 of 300 pairs, so single verdicts mean little. Categories rest on about 30 prompts and the persona on 24; a second seed or judge did not fit our budget of about 7 GPU hours.

Configs, metrics, data manifests, generations, pairs and judgments are under `configs/`, `outputs/` and `data/manifests/` as listed in the README; `scripts/make_tables.py` and `scripts/make_figures.py` rebuild every table and figure, and notebooks `00` to `06` rerun the experiments.